# Zero-Shot Cross-Modal Retrieval for Remote Sensing Images With Minimal Supervision

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***Abstract*— The performance of a deep-learning-based model primarily relies on the diversity and size of the training dataset. However, obtaining such a large amount of labeled data for practical remote sensing (RS) applications is expensive and labor-intensive. Training protocols have been previously proposed for few-shot learning (FSL) and zero-shot learning (ZSL). However, FSL is not compatible with handling unobserved class data at the inference phase, while ZSL requires many training samples of the seen classes. In this work, we propose a novel training protocol for image retrieval and name it as label-deficit zero-shot learning (LDZSL). We use this novel LDZSL training protocol for the challenging task of cross-sensor data retrieval in RS. This protocol uses very few labeled data samples of the seen classes during training and interprets unobserved class data samples at the inference phase. This strategy is critical as some data modalities are hard to annotate without domain experts. This work proposes a novel bilevel Siamese network to perform the LDZSL cross-sensor retrieval of multispectral and synthetic aperture radar (SAR) images. We use the available georeferenced SAR and multispectral data to domain align the embedding features of the two modalities. We experimentally demonstrate the proposed model's efficacy using the So2Sat dataset compared with the existing state-of-the-art models of the ZSL framework trained under a reduced training set. We also show the generalizability of the proposed model using a sketch-based image retrieval task. Experimental results on the Earth on the Canvas dataset exhibit comparative performance over the literature.**

***Index Terms*— Cross-modal retrieval (CMR), few-shot learning (FSL), multispectral, remote sensing (RS), synthetic aperture radar (SAR), zero-shot learning (ZSL).**

## I. Introduction

DUE to the availability of satellite data in abundance, content-based data retrieval is a critical problem statement in remote sensing (RS). With advancements in sensing technologies, many upcoming satellite missions can capture data using new-age sensors. The availability of various sensors necessitates developing cross-sensor/model retrieval frameworks as each sensor imagery captures different physical properties of complex land-use/land-cover patterns. For example, while a multispectral and hyperspectral satellite may help capture the amount of water and chlorophyll content to access plant health and vigor [1], a synthetic aperture radar (SAR) captures data at the microwave region of the electromagnetic spectrum which might be more robust in differentiating between phenologies of finer-grained crops such as rice and wheat-based from their backscatter pattern [2]. Therefore, each sensor can capture a specific subset of information from an area under observation. Leveraging the advantage of both the sensors might be essential in designing advanced agricultural monitoring applications in RS. We can consider each of these imaging sensors as a "modality." Hence, the paradigm of cross-modal retrieval (CMR) [3], [4] has attracted an increasing amount of research interest. The major challenge of CMR is to measure the similarity between two different types of data descriptions.

Furthermore, albeit CMR possesses a critical challenge that has started gaining attention, practical application models may often face two main challenges. One major drawback to this type of CMR problem is that typically they require an extensive database of annotated training samples. Getting that amount of labeled data in practical applications is expensive and labor-intensive. Some data modalities are especially difficult to annotate without domain experts, like SAR data, which gives us backscatter information at the microwave wavelength. Another major challenge with RS is that Earth observation often deals with previously unobserved phenomena. So the trained model encounters previously unseen or untrained class samples upon its deployment. In such a case, retraining the network with the newly discovered sample class may lead to catastrophic forgetting [5] of the learned weights.

These circumstances lead to the requirement of using either zero-shot learning (ZSL) [6] or few-shot learning (FSL) [7] techniques to handle either of one of the above-mentioned major challenges. Researchers have come up with the zero-shot learnable models for practical applications to deal with unobserved phenomena at the inference phase. A robust retrieval system that handles unseen class queries upon its deployment is called a zero-shot cross-modal retrieval (ZS-CMR) framework [8], [9]. Zero-shot retrieval mainly finds its application in problems where one may train the model on some set of seen classes. Upon deployment, it encounters previously totally unseen classes.

The fundamental idea behind designing an efficient zero-shot classification/retrieval network is to leverage the *semantic information* of the classes present in the dataset. Usually, the semantic information is chosen in such a way that it

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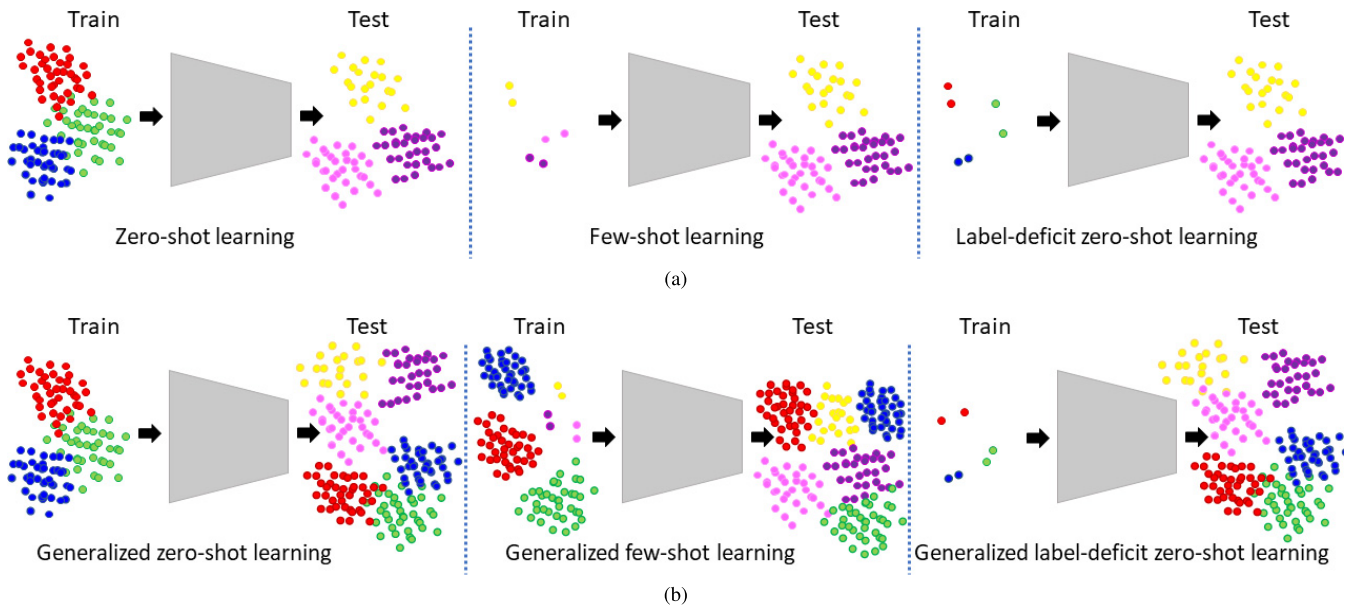



Fig. 1. Illustration of various types of (a) learning protocols and their (b) generalized learning protocols. The dots belonging to a specific color represent data (features) from a specific class. In the zero-shot case, the network is trained on a set of seen classes which is a complementary set of the unseen class samples that are fed to the network during the inference phase. A few-shot network comprises **at least** a few labeled samples of all the classes. Only the seen classes have a few labeled training samples in the label-deficit zero-shot case, and the model encounters both the sparingly seen and the unseen classes at the inference phase. In the generalized case, the model encounters the entire set of classes during the inference phase.

preserves the interclass relationship of the classes in the visual domain [10], [11]. To better elucidate this aspect, for example, the semantic vector describing a `broccoli` and a `cauliflower` should be nearer to each other in the Euclidean space than the semantic vectors of a `broccoli` and `building`. The main downside to this type of training protocol is the requirement of having a vast number of annotated samples of the seen classes to align the visual feature space to the semantic feature space.

An FSL protocol is usually preferred to tackle the issue concerning the availability of many labeled samples for training a model. FSL protocols are beneficial for training data that have an imbalance in sample distribution. Training a model with a large sample of some training classes, while only a few examples for the inference phase classes, is called FSL. Naturally occurring stretches of land-use/land-cover patterns are inherently imbalanced for any given region. A related example in this context is the fruit durian, native to Indonesia, Malaysia, the Philippines, and Thailand but is not grown anywhere else in the entire world. Most land-use and land-cover types, such as wheat or paddy fields, dominate over the rarely occurring ones. In most cases, one observes that the number of samples per class for naturally occurring scenes follows a *long-tail* distribution [12]. In such cases, it is noted that most standard classifiers perform reasonably well on the classes occurring around the head of the distribution, having the maximum number of training samples. However, they seem to underperform in learning the classifiers for the classes occurring around the tail of the distribution as they have very few training samples. In this regard, FSL helps in learning classifiers for the classes occurring at the tail of the distribution. Numerous methods have been proposed which use the FSL training protocol to tackle classification [13], domain adaptation [14], retrieval [15], and detection [16] tasks.

However, most studies on both the few-shot and ZSL algorithms are primarily based on computer vision applications and have not been adequately explored in RS. One can note that while few-shot retrieval has only been recently started to be explored in RS [17], [18], [19], [20], zero-shot retrieval still remains majorly unexplored [21], [22], [23], [24]. In [25], the authors propose using an unsupervised approach to address the problem that completely lacks labeled training data. However, this is often not the case in practical applications, and there could be a few labeled training samples available for a subset of classes.

Here the question arises: Can we have a unified framework to solve both the problems simultaneously? Although from the previous discussion it is clear that the current state-of-the-art (SOTA) methods attempt to solve these two paradigms as two separate issues. This challenging situation limits the overall performance of practical models in RS applications, primarily due to the lack of large annotated databases. In this article, we investigate and show that it is possible to fuse the problem domains into a single unified architecture to solve a CMR task with very few annotated training samples and be robust enough to retrieve unseen class query samples.

To this end, we develop a novel problem formulation for handling ZS-CMR under limited supervision and propose a model that solves this task. We define this problem as training the network with very few labeled images and testing the model on an unseen set of classes (samples). This work aims to achieve a model trained on only a few labeled samples and robust enough to retrieve unseen class queries. It is hereby referred to as the label-deficit zero-shot protocol. Similar to

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the zero-shot protocol, the semantic prototypes of each class are needed to align the semantic and the visual space of the embedding vectors. Furthermore, the domain gap between the two modalities needs to be bridged for a high-precision retrieval system.

We leverage the adequate availability of georeferenced SAR (Sentinel-1) and multispectral (Sentinel-2) data (without any class-wise label) for learning a domain-invariant feature space. There are mainly two types of learning protocols available in the literature till now, as explained above, namely, ZSL and FSL [12], [26]. Fig. 1(a) illustrates the experimental protocol of both these types, along with our proposed novel label-deficit zero-shot protocol. Fig. 1(b) shows the generalized learning protocols of the same. In the generalized case, the model encounters the entire set of classes during the inference phase.

Our key contributions in this work are listed as follows.

1) We identify an exigent problem of training deep learning models for handling classification and retrieval tasks under extremely scarce annotated labels. This article focuses on RS applications of cross-modal data retrieval under a reduced training set. We propose a novel experimental protocol of label-deficit zero-shot learning (LDZSL), which aims to bridge the heterogeneity gap between two different data modalities.
2) We propose a model to solve the above task using a two-stage training process. We first learn a pretext task to bridge the domain gap between the two modalities. This is followed by a fine-tuning task that uses very few labeled training data to inject the class information for the seen classes. By exploiting the semantic information, we aim to align the visual and the semantic space and leverage the topology of the semantic space to efficiently retrieve cross-domain unseen class samples at the inference phase. On account of this, we propose a bilevel Siamese network that solves the task of label-deficit zero-shot CMR. The details of the model are defined in Section III.
3) We perform extensive experiments on two different RS applications, i.e., SAR to multispectral and sketch to image retrieval. We conduct the experiments on two different datasets and benchmark the results against several baselines and literary works closest to the proposed problem formulation.

The SAR and multispectral data are the two chosen modalities for the first set of experiments. The image patches are acquired from the Sentinel-1 and -2 satellites [27]. Since there are no conventional pretraining models available for these types of RS data, we propose to exploit the georeference information associated with each SAR and multispectral image to make pairs and train a Siamese network (without the class labels) to make the model learn that two patches from different modalities represent the same information. It predominantly helps in bridging the domain gap between the two modalities. To demonstrate the generalizability of the proposed experimental protocol and framework, we also conduct experiments on a different task of LDZSL for sketch-based image retrieval (SBIR) on the Earth on Canvas [22] dataset. We describe the formulation of the problem statement and the methodology in Section II.

## II. Related Works

This section discusses a few prior related works in computer vision and RS. We divide this section into broadly three subsections, 1) CMR, 2) FSL, and 3) ZSL.

### A. Cross-Modal Retrieval (CMR)

Some of the early works in CMR were conducted for images to RGB depth estimation and retrieval [28]. Following this, many upcoming works aimed to bridge the feature gap between a 3-D shape and a 2-D image. A very widely researched area under CMR is retrieving image to text or text to images [29], [30], [31], [32], [33]. Most of these approaches adopt a hashing technique to increase the efficiency of the retrieval system. Recently, the application of image to audio samples' retrieval has also been researched after it was found to be helpful for visually challenged users [34], [35]. This was an exceptionally challenging task as the visual and auditory feature space semantics were significantly far apart.

In RS, one gradually recognized the need for a CMR system to be of pertinent interest as multiple satellites are capturing the same region of interest with diverse information. This was first recognized by [35], [36] where the authors tried to map the multispectral and panchromatic feature spaces in both [35], [36] and the audio-visual feature space in [34], [35]. Following this, many more research articles started coming up. O late, SBIR [22], [37] and image to text retrieval [38] tasks have also gathered attention in RS. There have also been studies that handled cross-resolution retrieval tasks [39] where the authors consider bands of different resolutions to be of different modalities. On a similar line, [40] propose a learnable joint spatial and spectral transformation model for high-resolution RS image retrieval. Sun *et al.* [41] fused additional multispectral images that were generated by a generative adversarial network to design a multispectral to SAR CMR framework. This technique helped eliminate the inherent spatial–spectral discrepancies between the two modalities.

### B. Few-Shot Learning (FSL)

While the problem areas discussed above are trained on a huge set of paired training data from multiple modalities, obtaining these many training data samples may often be expensive and labor-intensive. To this end, it is more practical to train models that are trainable using very few labeled training data. The existing FSL problems are mostly supervised learning problems. It has been mainly explored for image classification [42], knowledge transfer [43], image retrieval [44], and object recognition [7]. SBIR problems have also been explored where multiple image categories are trained using large-scale annotated samples. In contrast, the remaining classes are trained and tested using only a few annotated samples. Wang *et al.* [45] made an exhaustive survey of FSL tasks that are available in the literature.

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In RS, the paradigm of FSL is beneficial as annotating data often require domain experts. To this end, [44] proposed an image retrieval framework for RS under model-agnostic meta-learning (MAML) framework. In [19], the authors propose a stable prototypical network that classifies hyperspectral images using FSL. On a similar line, [46] proposed an architecture using 3-D convolutional neural networks (CNNs) to extract the spectral and spatial features to achieve the same task. The authors in [47] handle a much more challenging task of object detection in RS images using the same FSL strategy. This is more challenging as object detection tasks typically require training with significantly large annotated samples.

### C. Zero-Shot Learning (ZSL)

Extending on the problem of FSL, when there is an absolute scarcity of training samples and the model encounters a previously unseen class during the inference phase, it is known as the ZSL. ZSL uses auxiliary semantic information to aid the training process and facilitates the unseen class mapping at the testing phase. The existing research primarily aims to learn a mapping function between the visual space and the semantic space [23], [24]. A comprehensive evaluation of the ZSL strategy is presented in [48].

In RS, the ZSL protocol has very sparsely been explored. Few of the existing works mainly handle object recognition [49], scene classification [50], scene retrieval [22], and target recognition tasks [51]. The traditional fully supervised algorithms often fail in handling fine-grained object recognition tasks where the interclass variance is very low and the intraclass variance is very high.

Further, delving deeper into the problem of ZSL, a much more challenging is using ZSL for CMR setup. It is challenging tolearn the mapping between two utterly different data modalities. The zero-shot sketch-based image retrieval (SBIR) taskis a common problem statement that is often tackled using this training protocol. A sketched query is passed to retrieve the related image samples. This area has received a lot of research attention recently for natural images, with [52], [53], [54], [55] being some of the pioneering works in this field.

Chaudhuri *et al.* [22] performed SBIR under a ZSL protocol. On a similar line, [37], [56] also proposed solutions to the SBIR problem. However, [37], [56] were fully supervised SBIR problems with the same train and test classes. Fig. 2 illustrates the overall architecture of the proposed label-deficit zero-shot cross-modal retrieval (LDZS-CMR) framework for SAR to multispectral and vice versa retrieval.

## III. Methodology

### A. Problem Formulation

Let us denote the two modalities of data as $\mathcal{A}$ and $\mathcal{B}$. The goal of LDZS-CMR is to be able to retrieve similar images from domain $\mathcal{A}$ or $\mathcal{B}$, given a query in $\mathcal{B}$ or $\mathcal{A}$, respectively. In addition, we aim to train the model using very few annotated samples of some classes and expect the model to perform well on untrained/unseen classes during inference.

For a zero-shot framework, the dataset is partitioned into two sets of classes, i.e., *seen* and *unseen*. Let us denote

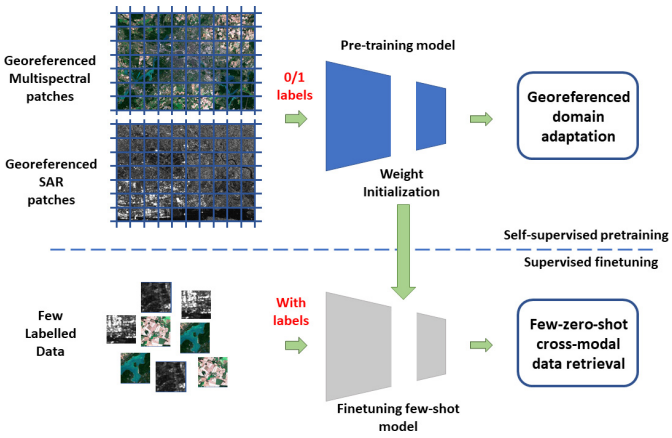



Fig. 2. Illustration of the overall architecture of the proposed LDZS-CMR framework for retrieving samples across the multispectral and SAR sensors. The pretext task uses the abundantly available georeferenced image pairs of SAR and multispectral data to bridge the domain gap between the embedding features. The fine-tuning network then uses very few labeled data of both the modalities to make the features class-wise discernible and perform the label-deficit zero-shot CMR task.

the class labels as $\mathcal{Y}$ and the semantic class prototype as $\mathcal{Z}$. To learn about the interclass relationship for the visual data, we borrow the semantic information from the natural language processing (NLP) space. For the semantic vectors, we exploited the standard `word2vec` [57] model to generate the word embeddings of the class names. Therefore, the seen training set and the unseen testing sets are represented as $\mathcal{S} = \{\mathcal{A}^s, \mathcal{B}^s, \mathcal{Y}^s, \mathcal{Z}^s\}$ and $\mathcal{U} = \{\mathcal{A}^u, \mathcal{B}^u, \mathcal{Y}^u, \mathcal{Z}^u\}$, respectively. Here, the strict condition is that $\mathcal{Y}^s \cap \mathcal{Y}^u = \emptyset$, meaning there should not be any overlap between the seen and unseen classes. Extending this to the LDZSL setup, we use any $k$ random samples from each class from the seen classes and use them for training the model. We exclude all the remaining annotated samples from the training data. Hence, we use only $k$-samples of $\mathcal{S}$ classes for training.

The overall goal is to bridge the domain gap between the features of both the data modalities in the embedding space and to design a mapping between the visual space and the semantic space to facilitate cross-domain data retrieval between $\mathcal{A}^u$ and $\mathcal{B}^u$. Here, we use a self-supervised module to pretrain the network by exploiting some of the existing information. In the case of sensor acquired data, we exploit the georeference information of the two different sensor modalities. We leverage the adequately available georeferenced Sentinel-1 and Sentinel-2 data (without any class-wise label) for learning a domain-invariant feature space. Likewise, in the case of SBIR, we use images used as a reference for drawing a sketch and pairing them together.

The pretraining network helps bridge the domain gap between the features of the two different sensor modalities. We aim to make the embedding features class-wise discriminative for the fine-tuning stage while learning a mapping function that maps the semantic space and the visual space to make the model generalizable for the unseen class data. In the generalized case, the model encounters the entire set of classes $\{\mathcal{A}^s \cup \mathcal{A}^u\}$ during the inference phase. The details about

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the proposed methodology are given in Section III-B. Fig. 3 illustrates the overall architecture of the proposed framework.

We divide the overall methodology broadly into two stages.

1) In stage I, we aim to bridge the heterogeneity gap [58]. To achieve this, we propose to exploit the georeference information to extract the features of the two modalities and pretrain the modality-specific encoder without knowing the actual class labels.

2) In stage II, we coconstruct features for both the modalities in the unified space by exploiting a Siamese model architecture. It synthesizes the shared space features of both the visual data modalities and the semantic prototypes for CMR. We perform training using a few samples at this stage and test the model on unseen classes.

### *B. Training Stage I: Georeferenced Domain Adaptation*

Standard RGB images having three channels, and incorporating natural image statistics [59] have the advantage of using an ImageNet pretrained encoder for initializing the weights. This could help to a great extent in enhancing the model's performance. In the case of RS images (Sentinel-1 and -2 in this case), except for unmanned aerial vehicle (UAV) captured optical images, most RS images have different bands, and the images are significantly different from the ImageNet classes. Hence, the ImageNet pretrained models such as ResNets [60] and VggNets [61] would not help much in these types of imageries.

In the absence of such a pretrained network, we exploit the georeference information. For any georeferenced patch of land scanned by both Sentinel-1 and Sentinel-2, we have a pair, and hence there is no need for any annotation of classes at this stage. Without loss of generality, one can state that for a given patch of land that has both overlapping Sentinel-1 and -2 data, they essentially constitute the same land-use/land-cover class. This work proposed to use this pair information in the pretraining stage to extract the preliminary level features and bring them close to each other in the embedding space for effective CMR.

A Siamese encoder–decoder framework is proposed that runs purely on pair information. Two patches from $\mathcal{A}$ and $\mathcal{B}$, belonging to the exact geolocation, are considered to form positive exemplars, while two patches from the two modalities, which are not from the precise geolocation, are regarded as the negative examples [62], [63]. The previous assumption is always valid. However, the second assumption regarding the negative exemplars may not always be correct if one has two patches from different geolocation of an extended patch. However, since many combinations of pairs are being considered, the probability of encountering such a positive pair combination while generating the negative examples would be considerably low. Hence, we proceed with these assumptions while training. Unlike conventional weight-sharing Siamese networks, since the two modalities of the proposed network are different, the model is allowed to learn two separate modality-specific encoders (without weight sharing).

Two encoders $f_A(;,\theta_{fA})$ and $f_B(;,\theta_{fB})$ using 2-D convolution layers (CNN) are used to encode the two input modalities. The CNNs are used in conjugation with spectral attention layers activated using sigmoid nonlinearity to highlight the most important features in the encoder stream. Here, $\theta_A$ and $\theta_B$ are the learnable encoder weights. Similarly, two modality-specific decoders, denoted by $g_A(;,\theta_{gA})$ and $g_B(;,\theta_{gB})$, are also used. Here again, $\theta_{gA}$ and $\theta_{gB}$ are the learnable decoder weights. The intermediate bottleneck layers of both the networks, $x_a$ and $x_b$, are used as inputs to the Siamese network. To this end, a few FC layers (FCN) are added to these bottleneck features $x_a$ and $x_b$, and these fully connected (FC) features are used for minimizing a contrastive loss function. The encoders and decoders would help the network learn robust representations for both the modalities. In contrast, the Siamese network will help bridge the domain gap between the feature embeddings of both the modalities. In contrast, the Siamese network will help bridge the domain gap between the feature embeddings of both the modalities. A contrastive loss is defined as follows:

$$\mathcal{L}_{\text{con}} = (1-y)(D_w)^2 + (y)\frac{1}{2}\max(0, m - D_w)^2. \tag{1}$$

Here, $D_w$ defines the mean square error distance between the final feature embeddings of both the modalities, i.e., $D_w = f_A(\mathcal{A}_i^s)^2 - f_B(\mathcal{B}_i^s)^2$. Here, $m$ defines the margin beyond which the embeddings of two negative exemplars are pushed apart from each other, and $y$ defines the paired label. For positive examples, $y$ is assigned as 0, and for negative exemplars, $y$ is assigned as 1. It can be seen from 1 that by assigning $y = 0$ to a pair of positive exemplar, the second term of $\mathcal{L}_{\text{con}}$ is essentially removed. Hence, the network minimizes the mean square distance by bringing the two embeddings closer to each other in the embedding space. By assigning $y = 1$ to a pair of negative exemplars, the first term of $\mathcal{L}_{\text{con}}$ is dropped off. The remaining loss function pushes the two features further apart in the embedding space by at least $m$.

Thus, the network effectively aims to bring two features of patches from the exact location acquired from different sensors overlapping in nature. This essentially comprises the pretext task of pretraining the network to learn robust representations from both the sensors. Since this network does not require manual data annotation, it can automatically be learned on any number of image pairs' samples from the existing data resources. The more data we use, the better the mapping would be. The network is not pretrained with any unseen class samples to maintain the zero-shot protocol.

### *C. Training Stage II: Training With Limited Supervision*

In the next stage, the previously saved modality-specific encoders $f_A(;,\theta_{fA})$ and $f_B(;,\theta_{fB})$ are reloaded. Here, the $k$-annotated samples from the seen classes $\mathcal{S}$ are used to fine-tune the network using the class labels. The semantic prototypes are also used to align the visual space and the semantic space for a robust retrieval performance for unseen class data.

*1) Bilevel Siamese Network:* The reloaded modality-specific feature encoders are followed by a few layers of 2-D convolutions. The features are enhanced by adding a spectral attention module in the new encoder stream $f'_A(;,\theta_{f'A})$ and $f'_B(;,\theta_{f'B})$. The last stream of CNN, followed by FCN,

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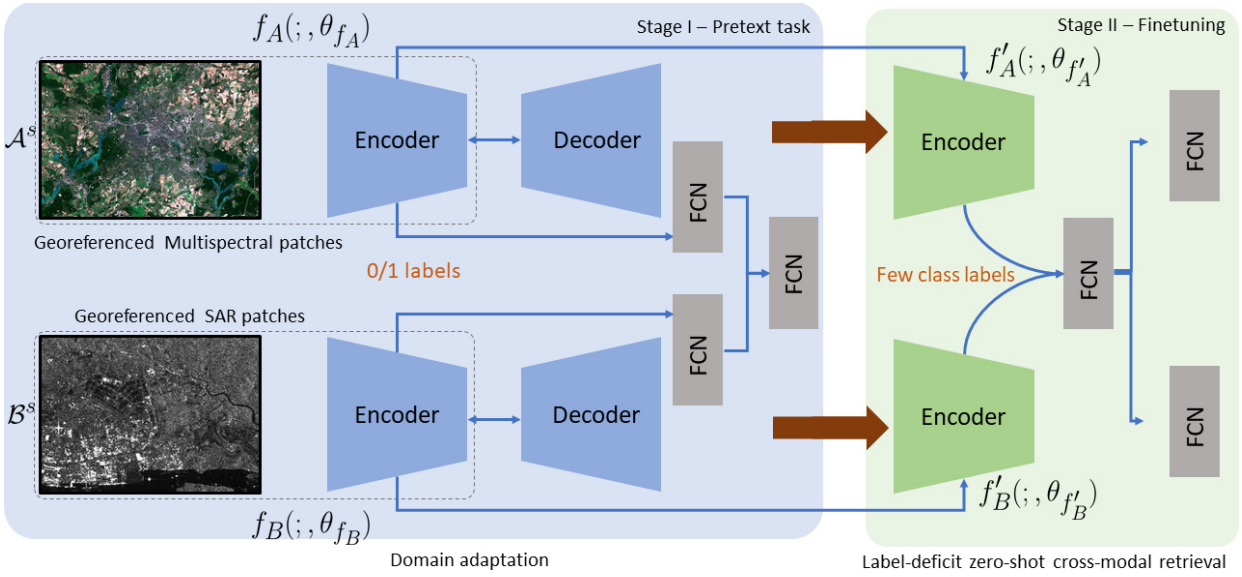



Fig. 3. Overall architecture of the pretraining stage and the fine-tuning stage. First, stage I is trained as the pretext task and the encoders $f_A(;,\theta_{f_A})$ and $f_B(;,\theta_{f_B})$ are learned from the georeference information. Patches from the same geographic location from the two different sensors are assigned the label 1, while from two different geographic locations are assigned 0. This helps in the initial bridging of the large domain gap across the sensors. In the next stage, the previous encoders are loaded and fine-tuned using few labeled training samples of SAR and multispectral to learn the encoders $f'_A(;,\theta_{f'_A})$ and $f'_B(;,\theta_{f'_B})$. This helps in forming class-wise discernible clusters.

shares their weights as by this stage, we have plugged in the necessary modules to bridge the domain gap between the two modalities. On the final FC layer, we apply another contrastive loss function. The positive and negative exemplars are created based on their class labels. Any pair of the image which shares the same label from $\mathcal{Y}^s$ is used as a positive example, while any two image pairs which do not have the same labels are assigned as the negative example. The number of annotated samples for training is very less, and hence to train the network, we used all the possible combinations of the samples

$$\mathcal{L}_{\text{con2}} = (1-y)\left(D'_w\right)^2 + (y)\frac{1}{2}\max\left(0, \phi - D'_w\right)^2. \quad (2)$$

Here, $D'_w$ defines the mean square error distance between the final feature embeddings of both the modalities of the fine-tuning stage, i.e., $D'_w = f'_A(\mathcal{A}^s_i)^2 - f'_B(\mathcal{B}^s_i)^2$. Here, $\phi$ defines the margin beyond which the embeddings of two negative exemplars are pushed apart from each other, and $y$ defines the paired label. For positive examples, $y$ is assigned as 0, and for negative exemplars, $y$ is assigned as 1. The network brings the two embeddings of the same class closer to each other in the embedding space by minimizing the mean square distance while simultaneously pushing the two embeddings of two different classes further apart in the embedding space by at least a margin of $m$.

*2) Visual–Semantic Space Alignment:* In the pretraining stage, we trained a Siamese network using georeferenced exemplars. Another Siamese network using the class labels of the few annotated images is trained in the next stage. Furthermore, to map the semantic space with the visual space, the semantic prototypes of the class labels from $\mathcal{Z}^s$ have been used

$$\mathcal{L}_{\text{sem}} = \frac{1}{n}\sum_{i=1}^{n}\left(f'_A\left(A^s_i\right) - Z^s\right)^2 + \frac{1}{n}\sum_{i=1}^{n}\left(f'_B\left(B^s_i\right) - Z^s\right)^2. \quad (3)$$

*3) Injecting Class-Wise Discernibility:* To this end, a cross-entropy (CE) classifier on generated semantic embeddings was used to condition the generated semantic embeddings better. The classifier is a two FC layer, with the final layer having softmax activation. The losses of semantic embedding and the classification task are balanced independently

$$\mathcal{L}_{\text{CE}} = \text{CE}\left(f'_A\left(A^s_i\right)\right) + \text{CE}\left(f'_B\left(B^s_i\right)\right). \quad (4)$$

Hence, we define the overall objective function as the linear combination of summation of all the three losses of the second stage, i.e., the contrastive loss, the semantic loss, and the label preserving cross-entropy loss. It is given in 5

$$\mathcal{L} = \lambda_1\mathcal{L}_{\text{sem}} + \lambda_2\mathcal{L}_{\text{CE}} + \lambda_3\mathcal{L}_{\text{con2}}. \quad (5)$$

Here, $\lambda_1$, $\lambda_2$, and $\lambda_3$ are the model hyperparameters. Since we do not know about the unseen classes that the model would encounter in the testing phase during the training process, one cannot use the conventional cross-validation method for tuning the hyperparameters. To accommodate this uncertainty of the vast possibilities of classes that the model may encounter at the inference phase, we divide the training classes into pseudo-seen (training) and pseudo-unseen (validation) classes. We then perform the standard cross-validation process to tune the hyperparameter values. We use this model on the full dataset, comprising the entire unseen test classes.

### D. Inference: Label-Deficit Zero-Shot CMR

Owing to the multiple loss functions in the network (Siamese and cross-entropy), training the model becomes

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**Algorithm 1** Training Phase of the Proposed LDZS-CMR Algorithm

**Input:** Georeferenced $\{\mathcal{A}^s, \mathcal{B}^s\}$, and $m$,
**Output:** Learned encoders $f_A(;,\theta_{fA})$ and $f_B(;,\theta_{fB})$.
Normalize $\mathcal{A}^s$ and $\mathcal{B}^s$.
$y = 0/1$ depending on geolocation of $\mathcal{A}_i^s$ and $\mathcal{B}_i^s$.
Define $D_w = f_A(\mathcal{A}_i^s)^2 - f_B(\mathcal{B}_i^s)^2$.
**do**

$$\min_{\theta_{fA},\theta_{fA}} (1-y)(D_w)^2 + (y)\frac{1}{2}\max(0, m - D_w)^2$$

**while** until convergence
**return** Pretext encoders: $f_A(;,\theta_{fA})$ and $f_B(;,\theta_{fB})$

**Stage II:**

**Input:** $k$-samples of $\{\mathcal{A}^s, \mathcal{B}^s, \mathcal{Y}^s, \mathcal{Z}^s\}$; $\lambda_1$, $\lambda_2$, $\lambda_3$ and $\phi$.
**Input:** Encoders $f_A(;,\theta_{fA})$ and $f_B(;,\theta_{fB})$
**Output:** Learned encoders $f'_A(;,\theta_{f'A})$ and $f'_B(;,\theta_{f'B})$.
$y = 0/1$ depending on $\mathcal{Y}^s$.
Generate $\mathcal{Z}^s$: `word2vec` encodings of class labels.
Define $D'_w = f'_A(\mathcal{A}_i^s)^2 - f'_B(\mathcal{B}_i^s)^2$.
**do**

$$\min_{\theta_{f'A},\theta_{f'B}} \lambda_1\mathcal{L}_{\text{sem}} + \lambda_2\mathcal{L}_{\text{CE}} + \lambda_3\mathcal{L}_{\text{con2}}$$

**while** until convergence
**return** Fine-tuned encoders: $f'_A(;,\theta_{f'A})$ and $f'_B(;,\theta_{f'B})$.

a nonconvex optimization problem. To this end, we have used the standard method of keeping one loss constant and minimizing the other at each step. We use the Adam optimizer with a stochastic mini-batch gradient descent approach to train the model. The trained model essentially consists of the embeddings of the modalities near-overlapping in the shared feature space while retaining its class-wise discriminative property. Once we obtained the trained model, we used it to generate the embeddings of the unseen samples for both the modalities. We used the $k$-nearest neighbor technique to retrieve cross-modal data instances, given the embedding of a query sample. We demonstrate the overall algorithmic steps for the training phase in Algorithm 1.

## IV. Experimental Setup

*Datasets:* For the first set of experiments, the SAR and multispectral data have been used. We demonstrate the generalizability of the proposed framework by conducting experiments for retrieving images from hand-drawn sketch queries.

### A. Multispectral and SAR Retrieval

We used the large-scale benchmarked So2Sat dataset [27] consisting of Sentinel-1 and -2 samples to test the proposed framework. The dataset consists of 400 673 pairs of the corresponding Sentinel-1 SAR and Sentinel-2 multispectral image patches with local climate zone (LCZ) labels. The images were collected across 52 major cities worldwide and ten additional cities. The classes were assigned labels based on the building surface fraction, previous surface fraction, impervious surface fraction, and height above ground (in meters). Seventeen broad classes were fetched using the attributes mentioned above, namely, Compact high-rise, Compact mid-rise, Compact low-rise, Open high-rise, Open mid-rise, Open low-rise, Lightweight low-rise, Large low-rise, Sparsely built, Heavy industry, Dense trees, Scattered tree, Bush, Low plants, Bare rock, Bare soil, and Water.

We chose 14 classes out of 17 for training, while we chose three classes for testing. In Section IV-B, the classes `heavy industry`, `bare rock`, and `water` have been used as the unseen classes. For pretraining the proposed network, we chose ≈600 pair of images per class (without their class labels). For the next stage, we used ten random pairs of annotated samples per class to train the remaining network. For evaluation, we consider all the samples of the three unseen classes from the test dataset.

### B. Sketch and Image Retrieval

Furthermore, to show the generalizability of the LDZSL experimental protocol and the proposed model, we also handle an LDZS-CMR task for the SBIR problem in RS. Here, we used the benchmarked aerial sketch-image dataset of the Earth on Canvas dataset [22]. It consists of photo images taken from a subset of classes from the UC Merced dataset [64], and the sketch images were handmade by amateur artists. This type of problem statement is essential when the notion of query image exists as a vague perception in the user's mind, and there are no actual query images at their disposal. This dataset consists of 14 classes of aerial sketch and photo images, broadly airplane, baseball diamond, buildings, freeway, golf course, harbor, intersection, mobile home park, overpass, parking lot, river, runway, storage tank, and tennis court. Out of these, we choose three fine-grained classes of intersection, overpass, and freeway as the unseen class data.

*1) Model Architecture:* For the pretraining stage, we took the last four bands of the SAR data of the So2Sat dataset for each patch. The input patches are of the dimension ($32 \times 32 \times$ channel). The channel value for Sentinel-1 is four, while for Sentinel-2 is ten. For the encoders, the input layer is followed by stacking 2-D convolutions of channel dimensions 256 and 512. These layers are followed by spectral attention (using Global max pooling and ReLU activation). The bottleneck layer is of dimension ($7 \times 7 \times 512$). We construct the decoder layer using deconvolution layers of channel dimensions 256, 128, and 4. A batch normalization and upsampling layers follow each deconvolution layer. Finally, we apply a Siamese network to the flattened features of both these inputs. The Siamese module consists of various FC layers of dimensions 4096, 2048, 1024, 512, 256, and 1. Fig. 3 illustrates the overall model architecture (including the pretraining and fine-tuning stages).

For the next stage, we used a feature encoder having inputs of dimension ($7 \times 7 \times 512$). This step is followed by stacking 2-D convolution layers of dimensions 1024 and 32. The first layer is followed by another spectral attention

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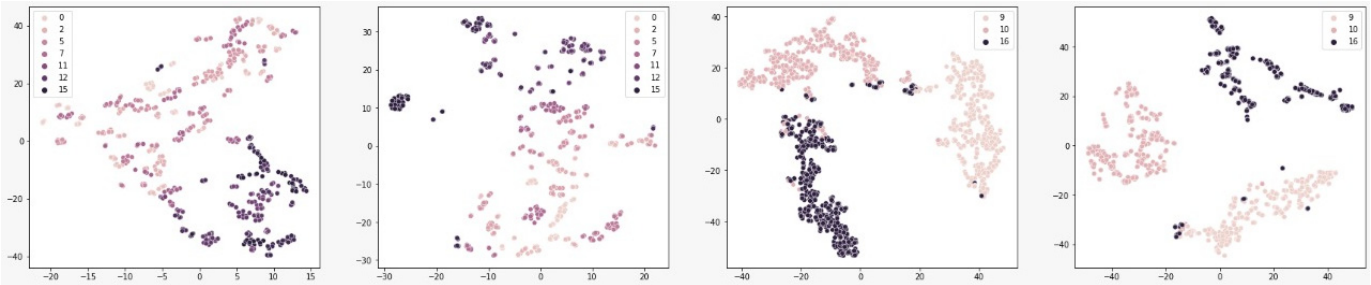


Fig. 4. 2-D scatter plots using $t$-SNE to show the embedding feature embeddings of the seen and unseen classes (unseen classes: `Sparsely built`, `heavy industry`, and `water`). The first two plots denote the t-SNE plots of the seen classes of Sentinel-1 and Sentinel-2, respectively. The next two plots denote the t-SNE plots of the unseen classes of Sentinel-1 and Sentinel-2, respectively. Each color denotes a separate class. Coarse-grained classes have been selected as the unseen classes in this set of experiments.

module which uses global max pooling and Sigmoid activation function. The final layer features of this second stage encoder give an output dimension of $(5 \times 5 \times 32)$. We append this layer with two more 2-D convolution layers and an FC layer. The last stream of CNN, followed by FCN, shares their weights as by this stage, we have plugged in the necessary modules to bridge the domain gap between the two modalities. On the final FC layer, we apply another contrastive loss function. This step is then followed by the Siamese network consisting of two 2-D convolutions of dimensions 128 and 256, which are then flattened, and subsequently, we apply an FC layer of dimension 512.

*2) Training and Evaluation Protocol:* The performance of the network is evaluated using mean average precision (mAP) and class-wise average precision values. The mAP of a set of queries is defined as $\text{mAP} = \sum_{q=1}^{Q} \text{Avg}(P(q))/Q$. Here, $Q$ is the number of queries in the set, and $\text{Avg}(P(q))$ is the average precision for a given query, $q$. We considered random pairs of image patches while training out of 8400 image pairs to train the first stage of the model. We carefully selected the data for the pretext stage to avoid having regions (classes) from the unseen classes. We choose a batch size of 32 and a learning rate of 0.001 to train the network. For the contrastive loss, we set the margin value to 1. At this stage, the Siamese network is trained purely based on georeferenced image pairs (without using their class labels), with no annotations.

For the second stage, we used a batch size of 32 and a learning rate of 0.001. For the semantic prototypes, we exploited the standard `word2vec` model to generate the word embeddings of the class names. Not all the combinations of words had feature embeddings in the `word2vec` space. So we took the average of the individual word embeddings of each class name. The second stage of the model is trained using randomly chosen ten sample pairs per class (using their class labels). Since the number of available samples is much less here to perform metric learning, we took all the possible combinations of pairs of patches together as the input to the model. For the contrastive loss, we set the margin value to 1.

## V. Results and Analysis

The proposed model is compared with the SOTA techniques of the Siamese network [65], Triplet network [66], Doodle2search [52], SEM-PCYC [54], and CrossAtNet [55]. While SEM-PCYC is based on adversarial training, Doodle2search and CrossAtNet use the triplet-based network backbone architecture and primarily solve a domain adaptation task. The Siamese network is created using a Siamese twin network, minimizing a contrastive loss [65].

TABLE I

Performance of the Proposed Cross-Modal Label-Deficit Zero-Shot Framework on the So2Sat Dataset in Terms of mAP Values

| **Task** | **Sen-1→Sen-2** | | **Sen-2→Sen-1** | |
|---|---|---|---|---|
| | 1-shot | 10-shot | 1-shot | 10-shot |
| Baseline | 28.76 | 30.26 | 20.77 | 22.45 |
| Siamese network [65] | 25.19 | 28.04 | 21.75 | 25.74 |
| Triplet network [66] | 28.98 | 30.91 | 25.87 | 28.77 |
| Doodle2search [52] | 37.76 | 52.00 | 29.67 | 32.00 |
| SEM-PCYC [54] | 31.65 | 38.35 | 34.88 | 37.26 |
| CrossAtNet [55] | 30.03 | 33.39 | 28.33 | 31.06 |
| **LDZS-CMR (Ours)** | **38.87** | **55.21** | **47.65** | **69.03** |
| **Generalized setup** | | | | |
| SEM-PCYC [54] | 10.21 | 10.37 | 13.43 | 16.00 |
| CrossAtNet [55] | 09.44 | 10.31 | 09.82 | 14.35 |
| **LDZS-CMR (Ours)** | **10.65** | **12.33** | **14.30** | **16.09** |

Since the network protocol deals with unseen image classes at the inference phase, an additional semantic loss has also been added to the final layer of the network to maintain fairness in comparison and to help align the visual space and semantic space well to help in the retrieval of the unseen classes. The triplet network is constructed using a simple triplet loss function [66] from the pretrained embeddings of stage 1. Again, we use a similar network comprising just the triplet loss and semantic losses to account for the unseen class sample upon the inference phase. In addition, we also show the model's performance on the baseline model, which has just the initial CNN extracted features.

It may be noted that this is a novel training protocol, and hence there is no available work in the literature to compare our work. Hence, the cross-modal ZSL frameworks trained for zero-shot SBIR tasks have been considered and are directly fed with the embeddings obtained from our pretrained network to maintain fairness and consistency in comparison. The second stage of the network is trained exactly as the model is defined but with only ten samples per class to maintain the LDZS-CMR protocol. Table I shows the performance of CMR for

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TABLE II

CLASS-WISE AVERAGE PRECISION FOR RETRIEVING UNSEEN CLASSES USING DOODLE2SEARCH (D2S) [52], SEM-PCYC [54], AND THE PROPOSED LDZS-CMR. TABLE SHOWS CROSS-MODAL AND UNIMODAL RETRIEVAL VALUES

| | Class | Sen-1→Sen-1 | | Sen-1→Sen-2 | | Sen-2→Sen-1 | | Sen-2→Sen-2 | |
|---|---|---|---|---|---|---|---|---|---|
| | | 1-shot | 10-shot | 1-shot | 10-shot | 1-shot | 10-shot | 1-shot | 10-shot |
| D2S | Heavy industry | 91.70 | 94.12 | 40.88 | 63.65 | 13.29 | 18.28 | 89.54 | 98.65 |
| | Bare rock | 79.77 | 82.31 | 71.90 | 93.11 | 47.59 | 66.04 | 88.00 | 98.49 |
| | Water | 79.79 | 88.00 | 0.63 | 0.0 | 10.22 | 14.82 | 95.45 | 99.01 |
| SEM-PCYC | Heavy industry | 90.44 | 92.92 | 47.17 | 93.17 | 21.04 | 28.93 | 88.67 | 98.08 |
| | Bare rock | 74.81 | 77.27 | 15.88 | 21.53 | 52.8 | 62.42 | 89.59 | 97.70 |
| | Water | 81.75 | 89.80 | 0.25 | 0.34 | 12.88 | 20.42 | 93.61 | 99.61 |
| Ours | Heavy industry | 87.45 | 90.58 | 78.65 | 85.65 | 42.56 | 88.18 | 91.08 | 98.52 |
| | Bare rock | 79.11 | 80.02 | 39.97 | 79.00 | 41.99 | 49.99 | 93.78 | 96.52 |
| | Water | 87.38 | 88.49 | 0.89 | 1.01 | 63.49 | 68.92 | 98.21 | 99.51 |

Sentinel (Sen) -1 to -2 and Sen-2 to -1. The values are reported in terms of mAP.

One can see from Table I that the proposed model clearly outperforms the existing SOTA models. While Doodle2search yields the best result for Sen-1 to -2 retrieval with an mAP value of 52.0, it still falls behind the proposed model with a margin $>3\%$. Likewise, we found the SEM-PCYC model to yield the best mAP of 37.26 for Sen-2 to -1 retrieval, and it falls grossly behind the proposed model, which gives an mAP of 69.03. In the generalized case where the model encounters the entire set of classes $\{\mathcal{A}^s \cup \mathcal{A}^u\}$ during the inference phase, the proposed model still outperforms the SOTA, resulting in an mAP value of 12.33 and 16.09 for the Sen-1 to -2 and Sen-2 to -1 retrieval, respectively.

The class-wise retrieval performance is also carefully studied for each of these models. Table II reports the class-wise average precision for retrieving unseen class samples using SEM-PCYC, Doodle2search, and the proposed model. One can observe that the performance of unimodal retrieval is much higher in each model compared with CMR.

Furthermore, one may also note that the class `water` performs poorly in almost all the models while performing CMR. This is primarily due to the hubness problem [67] which commonly occurs in zero-shot retrieval problems. This occurs when the feature embeddings of one class clutter the entire feature space and keep getting retrieved for most of the retrieved samples. For example, when the water class has somewhat occurred in one of the seen classes partially as a subregion. Possibly the water surface was very rough in the scale of the radar wavelength when the image was acquired. Therefore, the scattering characteristics could be confused with areas having multiple to random bounces of the scattered wave.

Thus, in such a case, the `water` class would be confused with those classes showing similar characteristics that the model has previously seen and would lead to erroneous retrieval. Fig. 4 illustrates the 2-D scatter plots using $t$-distributed stochastic neighbor embedding (SNE) to show the embedding feature embeddings of the seen and unseen classes (unseen classes: `Sparsely built`, `heavy industry`, and `water`). The first two plots denote the t-SNE plots of the seen classes of Sentinel-1 and -2, respectively. The next two plots denote the t-SNE plots of the unseen classes of Sentinel-1 and -2, respectively. One can see that the unseen classes are well-clustered; however, the seen classes still could be clustered in a better way. Coarse-grained classes have been selected as the unseen classes in this set of experiments. As the coarseness increases, the discernibility among various classes increases. Very fine-grained classes lead to the generation of cluttered clusters of composing classes.

TABLE III

PERFORMANCE OF THE PROPOSED UNIMODAL LABEL-DEFICIT ZERO-SHOT FRAMEWORK ON THE SO2SAT DATASET IN TERMS OF mAP VALUES

| Task | Sen-1→Sen-1 | | Sen-2→Sen-2 | |
|---|---|---|---|---|
| | 1-shot | 10-shot | 1-shot | 10-shot |
| Baseline | 73.50 | 78.43 | 78.45 | 82.70 |
| Siamese network [65] | 78.00 | 80.32 | 85.28 | 89.45 |
| Triplet network [66] | 80.22 | 82.33 | 87.92 | 91.34 |
| Doodle2search [52] | 82.76 | 88.00 | 92.11 | 98.00 |
| SEM-PCYC [54] | 82.19 | 86.66 | 91.37 | 98.46 |
| CrossAtNet [55] | 80.43 | 83.73 | 94.00 | 97.21 |
| LDZS-CMR (Ours) | 85.33 | 86.37 | 95.32 | 98.18 |

We also study the unimodal data retrieval performance obtained using the embedding features synthesized using the proposed model and the other SOTA models for the designed framework. One may note here that the main purpose of this work is to map the CMR, and hence the unimodal retrieval performances are not considered while training the network. Table III reports the unimodal retrieval performance for the unseen classes.

One can note that the SEM-PCYC model marginally outperforms the proposed LDZS-CMR by $\approx 0.3\%$. However, one may note from Table I that the SEM-PCYC falls over a margin

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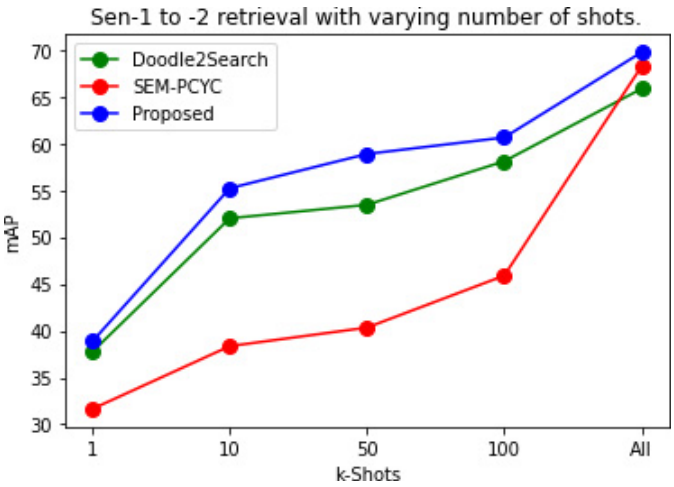



Fig. 5. Performance of Sen-1 to -2 retrieval with varying number of shots on Doodle2search [52], SEM-PCYC [54], and the proposed LDZS-CMR in terms of mAP values.

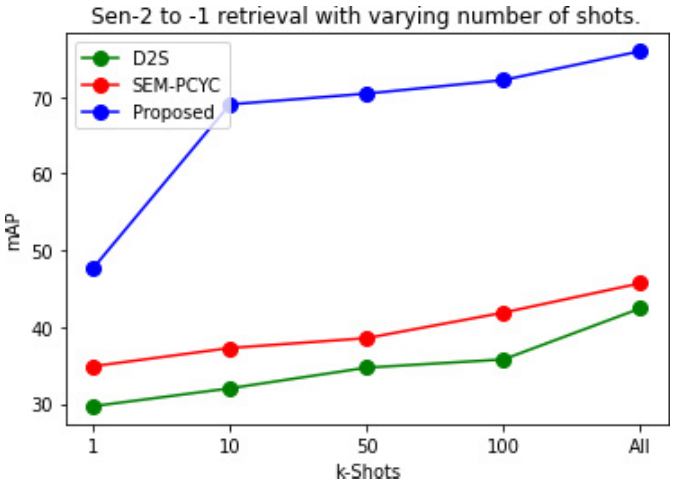



Fig. 6. Performance of Sen-2 to -1 retrieval with varying number of shots on Doodle2search [52], SEM-PCYC [54], and the proposed LDZS-CMR in terms of mAP values.

of ≈35% in the CMR of Sentinel-2 to Sentinel-1. This shows that the proposed LDZS-CMR framework is equally robust to retrieving unimodal retrieval. Furthermore, one may also note that unimodal retrieval is a relatively easier task to achieve, as seen from the performance than CMR.

### A. Effect of Varying the Number of Shots

To study the effect of varying the number of training samples, i.e., k-shots, we study the model's performance under different experimental setups. Doodle2search [52], SEM-PCYC [54], and the proposed LDZS-CMR have been trained with 1, 10, 50, 100, and all the samples individually, and their performance is shown in Figs. 5 and 6. It is seen that increasing the number of samples from 1 to 10 in all the models gives a considerable boost in the performance. From 10 to 100 is a marginal increase in all the three models, while adding all the samples from the training data yields boosts the performance. However, it is worth noting that the initial gradient from 1 to 10 is the maximum, and beyond that, the gradient change is only marginal. It could be worth pondering how much increase in the number of instances in the data is required given the marginal improvement that it causes.

### B. Effect of Semantic Information

While training the visual data is important, it is also essential that both the semantic and visual feature spaces are aligned well with each other. In the absence of a well-defined attribute vector (like the one used in [23]) which defines the properties of each of the seen and unseen classes, the semantic information is drawn from the word embeddings of the category names. There are various word embeddings that are available in the literature, for example, Word2vec [57], GloVe [68], Fasttext [69], to name a few. These are the distributed word-vector models pretrained on a large-scale database of WordNet. Since each of these models has been trained on a different experimental framework, their semantic feature space for each category name is oriented differently. In this section, we use these different word embeddings available in the literature and study the results (noted in Table IV). It is found that GloVe greatly underperforms by at least a margin of ≈5%–19% in comparison to the word2vec embeddings. On the other hand, Fasttext yielded better results than word2vec while retrieving Sentinel-1 to -2, but it performed poorly on the opposite domain query.

TABLE IV
EFFECTS OF DIFFERENT SEMANTIC INFORMATION ON THE mAP VALUE FOR LDZS-CMR RETRIEVAL ON THE SO2SAT DATASET

| W2v | GloVe | Fasttext | Sen-1 to -2 | Sen-2 to -1 |
|---|---|---|---|---|
| ✓ | | | 55.21 | 69.03 |
| | ✓ | | 51.37 | 50.33 |
| | | ✓ | 56.10 | 62.09 |

## VI. ABLATION STUDIES

In this section, we conduct several ablation studies of the proposed model. We ablate for the network components and for different training protocols. The ablation studies are performed on the So2Sat dataset, consisting of SAR and multispectral data.

### A. Ablation of Network Components

This section ablates the model by varying the network width and depth. We also study the contribution of each stage of the network.

*1) Effect of Pretext Task and Fine-Tuning Task:* To understand the contribution of each of the pretext and the fine-tuning model, we train the model on each of these components solely and check their results. We then compared the model performance on the complete network. First, we train the model using only the pretext task. At this stage, the model mainly helps bridge the domain gap between the two data modalities

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TABLE V
ABLATION OF VARIOUS NETWORK COMPONENTS IN TERMS OF mAP VALUES FOR TEN-SHOT LDZS-CMR

| Task | Sen 1→Sen 2 | Sen 2→Sen 1 |
|---|---|---|
| Pretext model | 34.46 | 62.53 |
| Fine-tuning model | 52.06 | 39.25 |
| Pretext + Fine-tuning model | **55.21** | **69.03** |
| Final FC layer width - 256-d | 46.16 | 31.08 |
| Final FC layer width - 128-d | **55.21** | **69.03** |
| Final FC layer width - 64-d | 51.71 | 30.87 |
| Final FC layer width - 32-d | 47.38 | 20.64 |
| FC network depth - 3 layers | 38.89 | 34.32 |
| FC network depth - 4 layers | **55.21** | **69.03** |
| FC network depth - 5 layers | 50.97 | 30.41 |

TABLE VI
ABLATION STUDIES OF DIFFERENT TRAINING PROTOCOLS

| Task | Sen 1→Sen 2 | Sen 2→Sen 1 |
|---|---|---|
| Few-shot learning | 52.06 | 39.25 |
| Zero-shot learning | 37.83 | 45.77 |
| Label-deficit zero-shot learning | 55.21 | 69.03 |
| 3 - Unseen classes | 55.21 | 69.03 |
| 4 - Unseen classes | 32.30 | 36.87 |
| 5 - Unseen classes | 22.62 | 30.95 |
| 6 - Unseen classes | 28.06 | 35.08 |
| All - Unseen classes | 34.46 | 62.53 |
| Course-grained unseen classes | 55.21 | 69.03 |
| Fine-grained unseen classes | 46.89 | 44.95 |

of SAR and multispectral data. One can see from Table V that the cross-domain retrieval performance of unseen data just using this stage itself has boosted the model performance considerably to an mAP value of 34.46 and 62.53 for SAR to multispectral and vice versa retrieval, respectively. This shows the effectiveness of the pretext layer, which is trained using just the georeference data and no labels at all.

When we consider just the fine-tuning layer, which is trained using only a few labeled image samples from both the domains, it is seen that the model is not able to generalize well over the vast range of unseen classes. Hence, the overall network performance is inferior. It is seen that the entire network, which is a combination of both the pretext and fine-tuning tasks, yields much more superior results. As seen from the $t$-SNE plots, the class-wise clusters of the Sentinel-1 are found to be relatively more dispersed than the clusters of Sentinel-2. Depending on the relative positioning of the chosen classes in the embedding feature space, one of the retrievals may dominate over the other one. This is the common observation in any form of cross-modal data retrieval.

*2) Effect of Network Width:* In the next set of experiments, we ablate for different network widths. When the penultimate layer of the network's width is changed, it is expected that the model performance will vary. As a rule of convention, we start using smaller layer width to collect as much local information as possible and gradually increase the layer width to reduce the generated feature space width to represent more global, high-level, and representative information. Too large layer width can lead to the problem of difficulty in converging to a minimal layer width in the final layer. Similarly, too small layer width also leads to difficulty in learning sufficiently robust representative features. Table V confirms the same upon experiments, and it is found that a 128-d layer width outperforms 32-d, 64-d, or 256-d widths.

*3) Effect of Network Depth:* It is commonly argued that one will learn better features as the network becomes deeper and deeper. However, as the network becomes too deep, the number of learnable parameters increases largely, affecting the overall network performance. As we deal with only $k$ training samples per class, going too deep may affect the network's overall performance. To this end, we experimented with three, four, and five layers before the final layer of the fine-tuning stage. It is seen that a depth of four layers yields the maximal mAP value of 55.21 and 69.03 for both the cross-modal unseen class data retrieval. The mAP values obtained by varying the depth of the network are shown in Table V.

### *B. Ablation of Different Training Protocol*

In this section, we ablate the model by varying the experimental setting. We also study the upper bound to the achievable results using a fully few-shot and a fully zero-shot training protocols.

*1) Effect of Varying the Training Protocol:* In this part of the ablation studies, we study the performance of few-shot and zero-shot experimental protocols on the proposed network and compare that with the performance of our proposed label-deficit zero-shot retrieval framework. Ideally, it is expected that few-shot and ZSL protocols requiring more annotated training samples should perform better than the proposed LDZSL as the latter works on very few labeled data points.

To this end, we train our model using the FSL protocol for the first set of experiments. In FSL, the model is trained on the entire training data of So2Sat for the 14 training classes and a few samples of the remaining three test classes. Since all the classes have at least a few annotated data samples, following the standard protocol of FSL [7], [70], we do not use the pretext model. Also, we discard the semantic loss since there is no unseen class. This yields an overall retrieval performance of 52.06 and 39.25 mAP for the cross-domain retrieval of Sen-1 to -2 and vice versa, respectively.

We train our model using the ZSL protocol for the next set of experiments. In ZSL, the model is trained on the entire training data of So2Sat for the 14 training classes and a zilch sample of the remaining three test classes. Since the seen classes have maximal annotated data samples and follow the standard protocol of ZSL [6], we do not use the pretext model. In this case, we use the semantic loss to align the visual space to the semantic feature space to generalize for the unseen test classes. This yields an overall retrieval performance of 37.83 and 45.77 mAP for the cross-domain retrieval of Sen-1 to -2 and vice versa.

There is a drop in the performance for zero-shot and few-shot compared with the LDZSL protocol, primarily attributed

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TABLE VII
PERFORMANCE OF THE PROPOSED CROSS- AND UNIMODAL LABEL-DEFICIT ZERO-SHOT FRAMEWORK ON THE EARTH ON CANVAS DATASET IN TERMS OF mAP VALUES

| Task | Image→Image | Sketch→Sketch | Image→Sketch | Sketch→Image |
|---|---|---|---|---|
| Siamese network [65] | 56.88 | 46.38 | 10.44 | 25.20 |
| Triplet network [66] | 64.08 | 51.61 | 12.43 | 17.87 |
| Doodle2search [52] | 77.00 | 77.00 | 36.00 | 58.00 |
| SEM-PCYC [54] | **82.29** | 82.34 | 36.60 | 54.75 |
| CrossAtNet [55] | 79.21 | 75.98 | 37.77 | 36.54 |
| **LDZS-CMR (Ours)** | 79.76 | **89.44** | **39.25** | **62.58** |

to the model architecture. In both the cases, we did not leverage the utility of the pretext model and instead relied more on the large-scale annotated data pairs ∼400 673. Furthermore, we also skipped the semantic embedding feature for the FSL experiments as there are no unseen data in this case. Hence, the effectiveness of the proposed model and the pretext model can be established here from the results obtained from the relatively more straightforward tasks of ZSL and FSL.

*2) Effect of Varying the Number of Unseen Classes:* Till now, all our experiments on the So2Sat dataset have a seen:unseen class split of 14:3. The three unseen classes chosen were `Heavy industry`, `Bare rock`, and `Water`. Now, suppose we increase the number of unseen classes. Then, in that case, we expect the following two phenomena likely to occur: 1) the CMR performance would start falling as lesser and lesser data clusters are annotated with their label tag. This would also lead to increased confusion among the existing classes and 2) as the unseen class increases, the number of seen classes decreases. This leads to having lesser confusion with the trained data classes, as shown in [71]. To investigate the exact trade-off, we gradually increase the number of unseen classes. We first add `open low-rise` buildings to the unseen category to get an mAP value of 32.30 and 36.87 for the two cross-modalities. We observe a considerable fall in the performance. We then add the `low plants` class to the unseen class bin to obtain a further fall of ≈6%–10% in the mAP values of both the cross-domain retrieval. We further go on to add another class as the unseen class. This time we add the `dense trees` class into the unseen class bin and test the CMR results. We again obtained inferior model performance compared with our initial experimental setup.

This set of experimental studies helps us confirm that the corresponding confusion with the previously seen class instances increases as the number of unseen classes increases. This leads to a fall in the network's overall performance, typical of any zero-shot problem. When all the 17 classes are referred to as the unseen classes (this is essentially just the pretraining stage), we found the CMR performance to be way better than the previous three, four, five, and six unseen class results obtained. This is so because there are no training classes to clutter and increase the confusion between the unseen classes. Hence, as expected from the concepts of ZSL [71], while too few seen classes lead to inducing confusion among the unseen classes during inference, not having any training class does not clutter the embedding feature space, leading to improved retrieval performance.

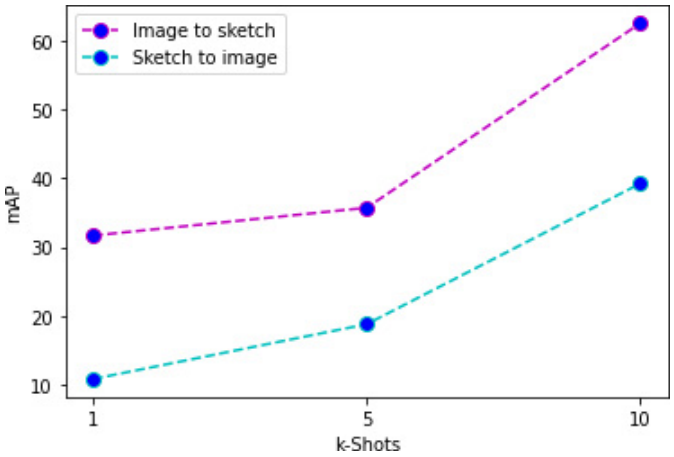



Fig. 7. Performance of image to sketch and sketch to image retrieval on the Earth on Canvas dataset with varying number of shots on Doodle2search [52], SEM-PCYC [54], and the proposed LDZS-CMR.

*3) Effect of Granularity of the Unseen Classes:* In the So2Sat dataset, some of the classes are very fine-grained, while some of the classes are course-grained. As the coarseness increases, the discernibility among various classes increases. Very fine-grained classes lead to the generation of cluttered clusters of composing classes. To study the effect of coarseness induced in the experiments, we keep course-grained classes as the unseen data in the first set of experiments. The unseen classes chosen here are `Heavy industry`, `bare rock`, and `water`. In the next set of experiments, we choose the `compact low rise`, `open low rise`, and `large low rise` as the fine-grained unseen classes. We observe a noticeable drop of ≈10% to 15% in both the cross-domain retrieval performance in terms of mAP values. Table VI reports the results obtained in both these experimental setups.

### C. Label-Deficit Zero-Shot SBIR

The proposed model is compared with the SOTA techniques of Doodle2search [52], SEM-PCYC [54], and CrossAtNet [55]. We also compare the performance with the previously elaborated baseline network and Siamese loss and triplet loss. Here, we choose to keep three fine-grained classes in the unseen phase. These are the `Intersection`, `Overpass`,

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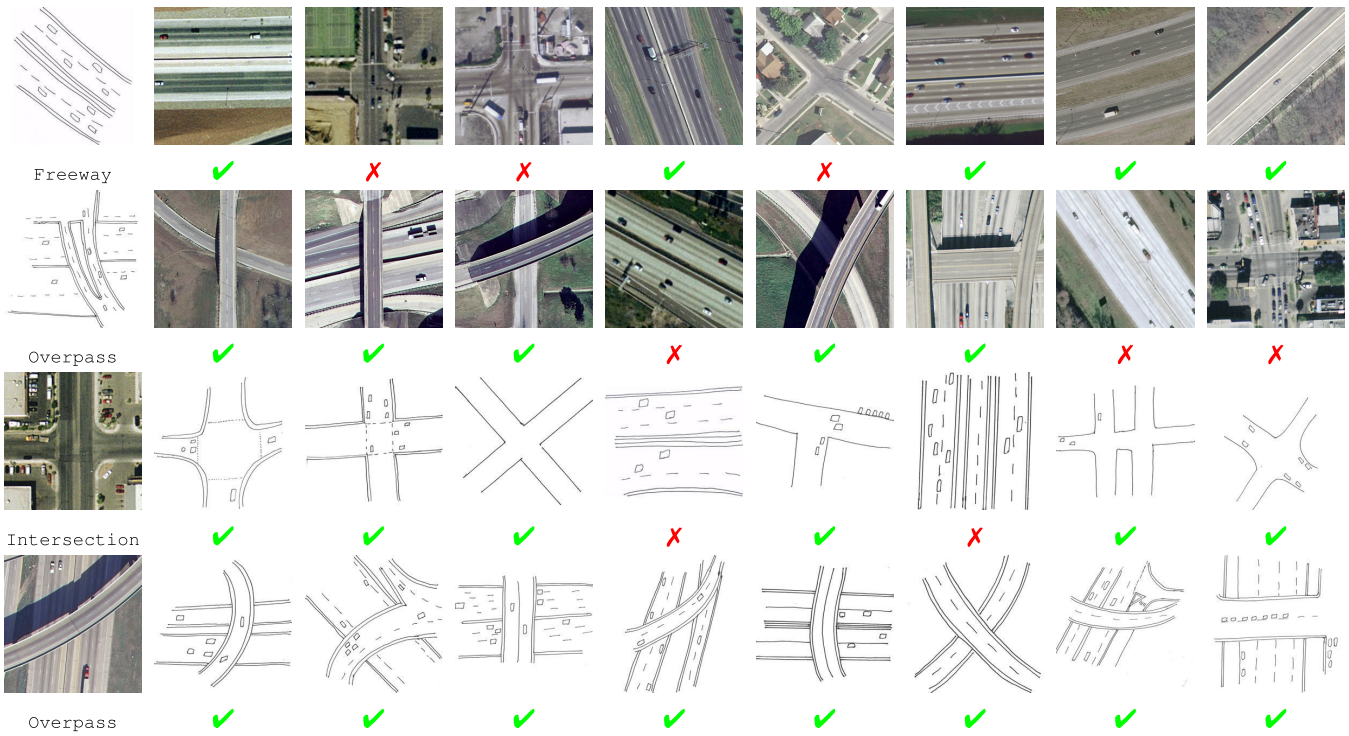



Fig. 8. Top-8 retrieval instances for a few sketch and photo queries from the EoC dataset. The green checks denote correctly retrieved classes, while the red crosses denote incorrect class images. The first column images correspond to the query picture.

and `Freeway` classes. It may be noted that this is a novel training protocol, and hence there is no exact available work in the literature to compare our work. Hence, the cross-modal ZSL frameworks trained for zero-shot SBIR tasks have been considered and are directly fed with the embeddings obtained from our pretrained network to maintain fairness and consistency in comparison. The second stage of the network is trained exactly as the model is defined but with only ten samples per class to maintain the LDZS-CMR protocol. Table VII shows the performance of CMR for an image to sketch and sketch to an image. It also reports the unimodal retrieval values of the sketch to sketch and image to image. We report all the values in terms of mAP, and we show the experiments for ten-shot LDZS-CMR.

In label-deficit zero-shot CMR, the SBIR works better than the image-based sketch retrieval performance. This is primarily because sketches consist of very less texture information, making it difficult for ImageNet pretrained CNNs to capture the features effectively as CNNs, which have a bias toward capturing textural properties over geometric spatial structures [72]. Sketches are entirely devoid of textures. From [72], we noted that increasing the shape bias improves the accuracy and robustness of a model. Therefore, to make the network learn both the SBIR and image-based sketch retrieval well, one would require a considerably bigger dataset to learn the pretext task.

However, given the performance of the SOTA SEM-PCYC and Doodle2search algorithms, it is seen that the proposed LDZS-CMR model outperforms both the CMR tasks with a considerable margin. On the other hand, the performance falls marginally in the case of unimodal image retrieval by a short margin of $\approx 2\%$. Fig. 7 illustrates the network performance by varying the number of shots from 1 to 10 for both the CMR setups for the proposed LDZS-CMR model. Finally, in Fig. 8 we show a few instances of retrieval images for both photo and sketch queries using the proposed LDZS-CMR framework.

## VII. Conclusion

The lack of annotated training samples is a vital issue in the deep-learning-based models for RS applications. It is well-known that creating a large-scale annotated database is both labor-intensive and expensive. Moreover, the RS models for various applications often encounter unobserved data instances during their inference phase. Therefore, retraining models from scratch for each new occurrence leads to catastrophic-forgetting of the learned model.

This work identifies these two cooccurring challenges and proposes a novel training protocol for LDZSL. The proposed protocol uses very few labeled data samples of the seen classes during training and is robust to handle unseen classes during its inference phase. We subsequently propose a novel bilevel Siamese network compliant with the LDZSL protocol. The model initially exploits georeference information of various satellite sensors for the pretraining stage. In the next stage, we fine-tune the model using very few labeled data samples.

We use the proposed training protocol and the proposed model to solve CMR tasks for two distinct types of data: 1) multispectral and SAR image retrieval and 2) sketch-image retrieval. The performance of the proposed model is then evaluated by training the model on very few samples of a subset of the class of the large-scale So2Sat and the Earth on Canvas datasets. One can clearly observe that the

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proposed network outperforms the SOTA model for both the LDZS-CMR and generalized LDZS-CMR tasks. In the future, we plan to take up the challenge of identifying small hubs [67] in the model, which clutters the retrieval space to perform better. A bilevel domain adaptation can be deployed on these models to overcome the hubness problem [73]. A possible extension of this problem can be active learning by keeping a human-in-the-loop. This can help rectify the incorrectly retrieved samples at the inference phase and reinforce the model with the added knowledge. We plan to look into these avenues in the future and develop a better and more robust framework.

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